

Chetankumar Khadke

New Jersey, USA

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SKILL SETS

Area:

Large Language Model
Agentic AI
Natural Language processing
Sentiment Analysis
Text Mining/Classification
Machine Learning
Deep Learning

Language:

Python, R, Java



[Chetan Khadke](#)



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PROFESSIONAL PROFILE

- Data Science Specialist with over 12 years of experience, specializing in Natural Language Processing (NLP) and Machine Learning (ML) for solving diverse and complex challenges across industries.
- Hands-on experience driving AI/ML solutions from Proof of Concept (POC) to production, translating client requirements into scalable, reliable, and production-ready solutions.
- Independently developed and deployed multiple Proof-of-Concepts (POCs) showcasing scalable NLP solutions, including integration with cloud platforms (AWS, GCP) and containerization for seamless deployment.
- Proficient in Python, R, and Java, with recent experience in using modern ML libraries like Hugging Face Transformers, LLM, and vision LLM to tackle NLP tasks.
- Published and presented four research papers in prominent international conferences, sharing novel insights and methodologies in NLP and ML applications.

PACKAGES/TOOLS: AI AGENT, HARNESS, AI FINE TUNING, LLAMA-INDEX, LANGCHAIN, GEMINI 3 PRO AND FLASH, AWS SONNET 5, LLAMA-CPP, LAYOUTLM, QWEN LM, DONUT, OPENAI, MLFLOW, BERT, PYTORCH

MORE SKILL SETS

Database: MySQL, MongoDB, Neo4j, Elastic Search

NLP: Large Language model (LLM)(Llama 3.2, Gemini 3 pro, AWS sonnet5, Mistral AI), Visual LLM (Llava, Qwen 2.5 VL), LayoutLM, Donut, UDOP, Text Classification and Clustering, Named Entity Recognition (NER), Entity Linking, Topic Modeling, Named Entity Linking

Agentic AI: AI Agents, Multi-Agent Systems, Agent Orchestration, Planning & Reasoning, Tool/Function Calling, Memory Management, Agentic RAG, Human-in-the-Loop, Model Context Protocol (MCP), Agent-to-Agent (A2A) Communication

AWS: ECR, ECS, S3, CloudWatch, SageMaker, Inference Endpoints, AWS Bedrock

ML Deployment: Docker container, Managing docker container with ECR, Git, ML flow

PROFESSIONAL EXPERIENCE

Xpanse

Staff Software Engineer
(Feb 2025 – Aug 2026)

Information Extraction from Documents – Xpanse, USA

Document AI is a highly scalable platform for automated document classification and data extraction. Out of the box it recognizes more than 500 document categories and captures over 500 distinct data elements, with customizable templates to onboard new layouts quickly. The system processes millions of documents each week, extracting not only text but also checkboxes, radio buttons, addresses, and signatures. Tightly integrated with a rules engine,

Document AI drives end-to-end automation across complex document workflows, dramatically reducing manual effort and cycle times.

(Feb 2025 – Aug 2026)

Responsibility:

- Architected and Engineered and fine-tuned prompt libraries with Gemini Pro 3.0, Flash, and AWS Anthropic Sonnet 5, delivering highly precise, context-rich information extraction.
- Introduced context-aware guardrails that sharply reduce hallucination rates and ensure consistent, trustworthy outputs.
- Built an automated framework that generates field-specific prompts on demand from document-field mappings, speeding the onboarding of new document types.
- Embedded the LLM pipeline into the production application, boosting overall extraction accuracy by up to ~ 30 percent.

Xoriant Solutions

Software Engineer

Information Extraction and Reconciliation – Xoriant, USA

The solution is dedicated to extracting critical information with utmost accuracy from trade documents sourced from various channels, including contracts, invoices, bills of lading, and financial statements. Designed to ensure data consistency, this solution performs robust reconciliation across multiple sources, identifying and reporting discrepancies in real time to mitigate potential financial losses. The solution integrates seamlessly into financial workflows, allowing stakeholders to address discrepancies proactively and make data-driven decisions with confidence.

Responsibility:

- Created and curated prompts leveraging the latest Gemini Pro 1.5 for optimized information extraction, ensuring high precision and relevance.
- Developed strategies to mitigate hallucination risks by crafting intelligent, context-aware prompts for reliable output.
- Designed advanced reconciliation status checks using Large Language Models (LLMs) and sentence-matching algorithms to improve data accuracy and consistency.
- Conducted defect validation and implemented bug fixes to enhance system stability and performance.

Information Extraction from Documents (Document AI) – Xoriant, Pune

Document AI is a robust, scalable solution for automatic document type detection and field/entity extraction. Capable of classifying over 500+ document types and extracting more than 500+ unique fields, Document AI allows the addition of custom templates to accommodate diverse formats. It efficiently processes millions of documents weekly, supporting not only text fields but also checkbox and radio button detection, address parsing, and signature recognition. Integrated with a rules engine, Document AI streamlines business processes by automating comprehensive document handling tasks.

Responsibility:

- Spearheaded the development of a Named Entity Recognition (NER) module, leveraging a variety of NER libraries to optimize performance and accuracy.
- Actively engaged in exploring and setting up client-centric annotation tools, tailoring solutions to meet specific client requirements.
- Led the design and training of customized models based on layoutLM, Donut and UDOP frameworks, enhancing their efficiency and applicability
- Successfully implemented a Donut-based DocVQA and LLM based approach, demonstrating proficiency in advanced document understanding techniques
- Achieved a remarkable 70% improvement in data extraction accuracy 3 through the implementation of innovative, custom-tailored solutions.
- Managed both the coding and deployment pipelines, ensuring seamless integration and functionality of machine learning models

(May 2021 –July 2024)

Technology Specialist

- Expertly deployed machine learning models in containerized environments, employing best practices for scalability and maintenance

Persistent Systems
(Apr 2019- May 2021)
Engineering Lead

Knowledge Graph – PSL, Pune

Created comprehensive knowledge graphs to establish connections between organizations, individuals, and content from diverse data sources, enabling enhanced data insights and relationship discovery

Responsibility:

- Conducted experiments with various Named Entity Recognition (NER) libraries to optimize entity extraction performance.
- Designed and implemented classification models for advanced relation extraction tasks, enhancing NLP model capabilities.
- Developed custom NER solutions tailored to project-specific requirements, improving accuracy and relevance in entity recognition.
- Executed semantic matching algorithms to enhance contextual understanding and similarity assessment within data.
- Performed Name Entity Linking (NEL) using SpaCy and Wikipedia resources to create robust entity relationships and improve model precision.
- Managed and optimized database operations with Neo4j, ensuring efficient data retrieval and storage for NLP applications.
- Deployed machine learning models in containerized environments to streamline scalability and integration within production workflow.

EDUCATION

(2009 – 2013)

Dharmsinh Desai University, Nadiad, Gujarat

Bachelor of Technology in computer science with GPA: 8.5/10, 2009-2013

ACHIEVEMENTS

- I have presented my work at the 11th International Conference on Evaluation of Novel Approaches to Software Engineering on 27-28 April 2016, Rome Italy.
- Bravo Award was received 3 times for Excellent work in Persistent System.
- XOR-Champ Award was received 1 time for Excellent work in Xoriant Solutions.

CERTIFICATIONS

- “Machine Learning” – Secured silver medal in course from Indian Institute of Technology – Madras (IIT-M) offered from NPTEL.
- “Deep Learning” – Indian Institute of Technology – Madras (IIT-M) offered from NPTEL.

RESEARCH PAPER DETAILS ^[1]

2015

1. Chetan Khadke, Sunjit Rana and Vipul Shah “Recovering Workflows from Functional Tests” in 23rd IEEE International Conference on Program Comprehension at Florence Italy.
2. Vipul Shah, Chetan Khadke and Sunjit Rana “Mining process models and architectural components from test cases” Software Testing, Verification and Validation Workshops (ICSTW), 2015 IEEE Eighth International Conference on 13-17 April 2015 at graz.
3. Vipul Shah, Chetan Khadke and Sunjit Rana “Knowledge Transition: Discovering models from functional tests” in the 26th IEEE International Symposium on Software Reliability Engineering on 2-5 November 2015 at GAITHERSBURG, MD, USA.

2016

4. Presented tool demonstration “Knowledge Acquisition (KnowX)”, tool demonstration in the 11th International Conference on Evaluation of Novel Approaches to Software Engineering on 27-28 April 2016, Rome Italy.

BLOG/PAPER LINK

1. Paper link: <http://dblp.uni-trier.de/pers/hd/k/Khadke:Chetan>
2. Medium blogs: https://github.com/khadkechetan/khadkechetan?tab=readme-ov-file#open_book-articles